

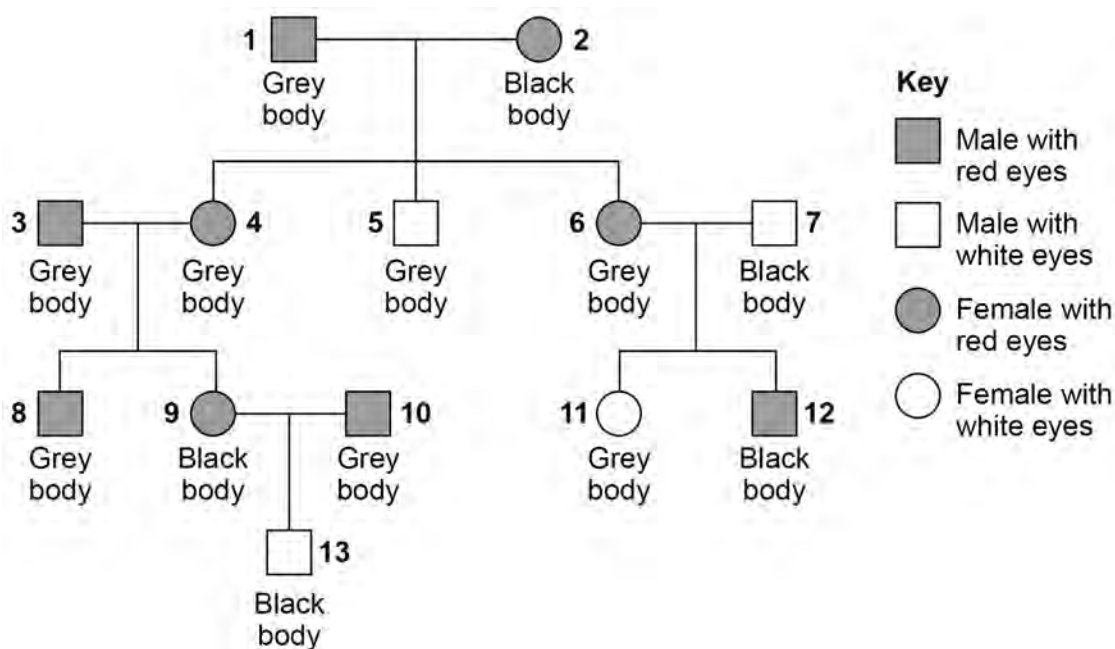
06

In fruit flies, a gene for body colour has a dominant allele for grey body, **G**, and a recessive allele for black body, **g**.

A gene for eye colour has a dominant allele for red eyes, **R**, and a recessive allele for white eyes, **r**, and is located on the **X chromosome**.

Figure 4 shows the phenotypes of fruit flies over four generations.

Figure 4



06.1

Give the full genotype of the fly numbered **6** in **Figure 4**.

[1 mark]

Genotype = _____

06.2

Give **one** piece of evidence from **Figure 4** to show that the allele for grey body colour is dominant.

[1 mark]

0 6 . 3

Explain one piece of evidence from **Figure 4** to show that the gene for body colour is **not** on the **X chromosome**.

[2 marks]

0 6 . 4

A heterozygous grey-bodied, white-eyed female fly was crossed with a black-bodied, red-eyed male fly.

Complete the genetic diagram below to show all the possible genotypes and the ratio of phenotypes expected in the offspring from this cross.

[3 marks]

Phenotypes of parents: Grey-bodied, white-eyed female × Black-bodied, red-eyed male

Genotypes of parents: _____ × _____

Genotypes of offspring _____

Phenotypes of offspring _____

Ratio of phenotypes _____

Question 6 continues on the next page

Turn over ►

0 6 . 5

A population of fruit flies contained 64% grey-bodied flies. Use the Hardy–Weinberg equation to calculate the percentage of flies heterozygous for gene **G**.

[2 marks]

Answer = _____ %

9

